DATA ITEM DESCRIPTION

Form Approved
OMB No. 0704-0188

2. TITLE	1. IDENTIFICATION NUMBER
Heat Balance Calculations and Flow Diagrams for Main Propulsion Plant Systems (Steam Driven Ships)	DI-MISC-80853

3. DESCRIPTION/PURPOSE

3.1 Steam plant design heat balances provide information related to the entire machinery plant system steam cycle to determine the following:

3.1.1 Proper rating of components, piping, and propulsion auxiliaries.
3.1.2 Anticipated operating conditions in the various systems of the propulsion plant.
3.1.3 Anticipated fuel rate in pounds per shaft horsepower-hour for non-nuclear ships.

4. APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC APPLICABLE	6b. GIDEP APPLICABLE
890614	NAVSEA 56X11		

7. APPLICATION/INTERRELATIONSHIP

7.1 This DID contains the format and content preparation instructions for the data product generated by the specific and discrete task requirement as delineated in the contract

7.2 These data apply only to steam driven ships and to the main machinery plants thereof. They do not apply to independent auxiliary machinery systems such as an auxiliary oil-fired or waste heat boiler steam heating system etc.

7.3 For non-nuclear ships, the heat balances provide fuel rates required for endurance calculations.

7.4 This DID supersedes UDI-E-23108.

8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		N4749

10 PREPARATION INSTRUCTIONS

10.1 Format. Design heat balances may be prepared in contractor format.

10.2 Content. The design heat balances, at the conditions specified in the ship specification, shall indicate:

10.2.1 The quantity, temperature, pressure and enthalpy of steam at appropriate locations.

10.2.2 The quantity and temperature of feed water and condensate entering and leaving each unit.

10.2.3 Where continuous bleeding traps are provided, the value of their total steam flow shall be indicated on the heat balance diagrams and taken into account in the heat balance calculations.

10.2.4 Design heat balances shall include calculations, flow diagrams, component performance curves, and a summary of assumptions made with appropriate justification remarks.

11. DISTRIBUTION STATEMENT

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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